Bright Learn

Machine Learning
Research Assignment

Research report | Parts A–D

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Machine Learning Research Assignment.

A1. What Is Machine Learning?

Machine learning is a branch of artificial intelligence (AI) that allows computers to learn patterns from data and make predictions or decisions without being given specific instructions for every task.

In traditional programming, a programmer writes rules that tell the computer exactly what to do. In machine learning, the computer learns patterns from examples and uses those patterns to make predictions about new data.

Example:

In traditional programming, a programmer can create rules to identify spam emails by checking for certain words.

In machine learning, a computer is trained using examples of spam and non-spam emails. It learns the patterns and uses them to identify whether a new email is spam.

Simple comparison:

Traditional Programming:

Data + Rules → Computer → Results

Machine Learning:

Data + Known Answers → Learning Algorithm → Trained Model

The trained model can then make predictions on new data.

A2. Artificial Intelligence, Machine Learning, Deep Learning and Data Science

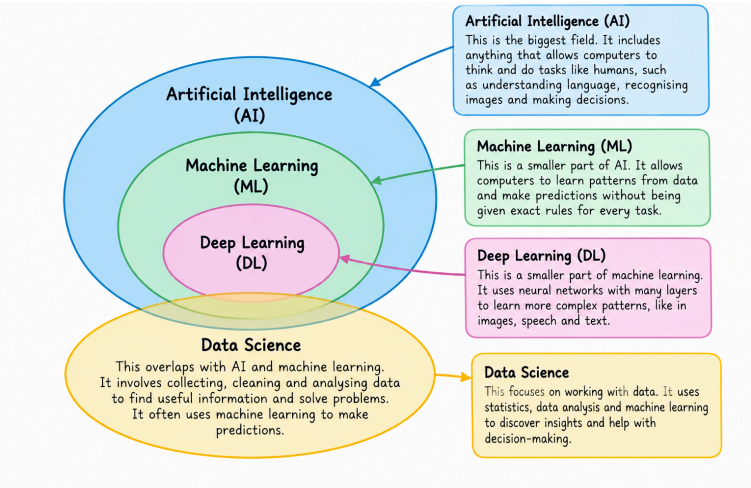
Artificial Intelligence (AI): AI is technology that allows computers to perform tasks that normally require human intelligence, such as understanding language and making decisions.

Machine Learning (ML): Machine learning is a part of AI that allows computers to learn patterns from data and make predictions without being programmed with every rule.

Deep Learning (DL): Deep learning is a part of machine learning that uses artificial neural networks with multiple layers to learn complex patterns. It is commonly used in facial recognition and speech recognition.

Data Science: Data science involves collecting, cleaning, analysing and interpreting data to find useful information. It uses statistics, programming and machine learning to solve problems.

Relationship: Artificial intelligence includes machine learning, and machine learning includes deep learning. Data science overlaps with these fields because it uses machine learning to analyze data and make predictions.



A3. Core Machine Learning Terms

1. Dataset: A dataset is a collection of information organized in rows and columns. For example, a student dataset may contain study hours, attendance and exam marks.

2. Feature: A feature is the information a machine learning model uses to make predictions. For example, study hours and attendance can be features used to predict a student's exam mark.

3. Label (Target): A label is the answer or result that the model is trained to predict. For example, the student's final exam mark is the target.

4. Model: A model is a system that learns patterns from data and uses them to make predictions. For example, a model can learn how study hours and attendance relate to exam marks.

5. Training: Training is the process of teaching a machine learning model using existing data. For example, we can train a model using students' study hours, attendance and their actual exam marks.

6. Prediction: A prediction is the result a trained model gives when it receives new data. For example, a model can predict a student's exam mark based on their study hours and attendance.

Example: Student Exam Marks Dataset

Study Hours	Attendance (%)	Exam Marks (%)
2	60	45
4	75	58
6	85	72
8	95	88

A4. Three Real-World Uses of Machine Learning in Africa

1. Fraud Detection in Banking

Banks in South Africa use machine learning to help identify suspicious transactions and prevent fraud.

- **Data used:** Transaction amounts, locations, times and customer spending patterns.
- **Prediction:** The model predicts whether a transaction might be fraudulent.
- **Who benefits:** Banks and customers benefit because it helps protect money and reduce financial losses.

2. Credit Scoring

Financial institutions use machine learning to help decide whether a customer is likely to repay a loan.

- **Data used:** Income, credit history, existing debts and repayment records.
- **Prediction:** The model predicts the likelihood of a customer repaying a loan or missing payments.
- **Who benefits:** Banks and customers benefit because it supports better lending decisions.

3. Crop Disease Detection in Agriculture

Machine learning can help farmers in African countries identify crop diseases early.

- **Data used:** Pictures of crops, weather conditions and information about plant health.
- **Prediction:** The model predicts whether a crop may have a disease.
- **Who benefits:** Farmers benefit because they can identify problems earlier, protect their crops and reduce losses.

A5. Machine Learning Workflow

The machine learning workflow is a series of steps followed when building a model to make predictions.

1. Problem Definition: First, we identify the problem we want to solve. For example, predicting whether a student will pass or fail an exam.

2. Data Collection: We collect the information needed to solve the problem, such as students' study hours, attendance and exam results.

3. Data Preparation: We clean the data by fixing missing values, removing duplicates and correcting errors so the model can learn properly.

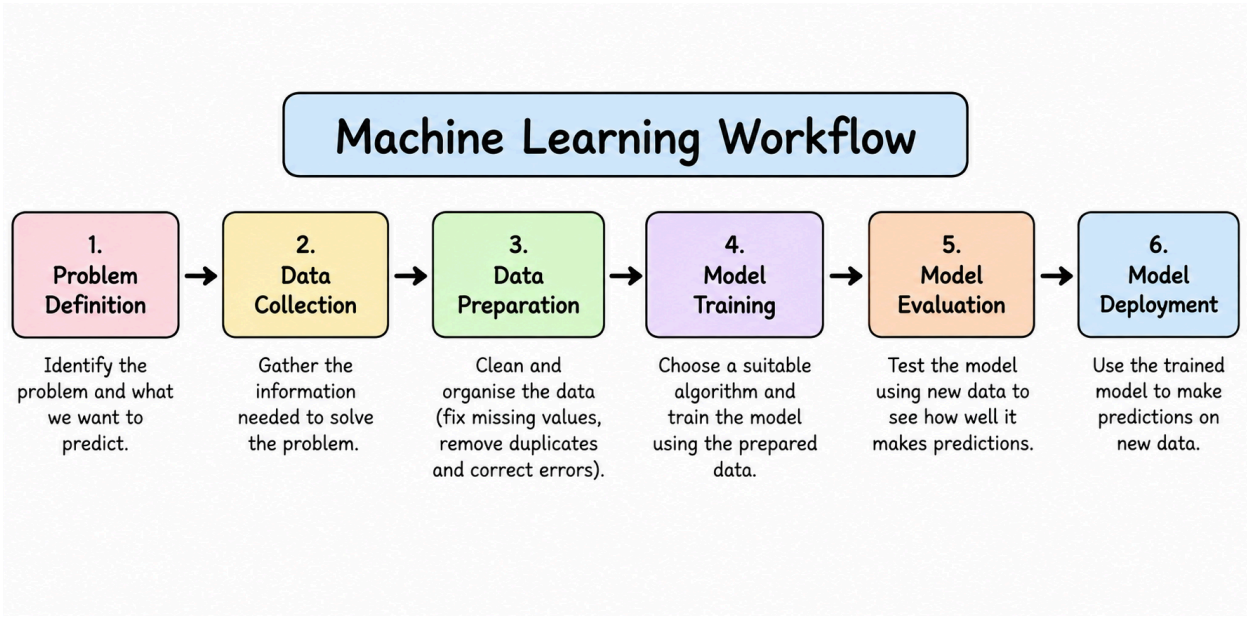
4. Model Selection and Training: We choose a suitable machine learning algorithm and train it using the prepared data.

5. Model Evaluation: We test the model using data it has not seen during training to check how well it makes predictions.

6. Model Deployment: Once the model performs well enough, we can use it to make predictions on new data.

Why does data preparation take the most time?

Data preparation usually takes the most time because real-world datasets often contain missing values, duplicates, incorrect information and different data formats. These problems must be fixed before training a model to improve the quality of its predictions.



B1. What Is Supervised Learning?

Supervised learning is a type of machine learning where a computer learns from data that already has the correct answers, called labelled data.

The model uses these examples to learn patterns so it can make predictions on new data.

Example:

A bank can use supervised learning to predict whether a customer will repay a loan. The model learns from previous customer records showing who repaid their loans and who did not. It then uses this information to predict whether a new customer is likely to repay a loan.

B2. What Is Unsupervised Learning?

Unsupervised learning is a type of machine learning where a computer learns from data that does not have correct answers or labels.

The model looks for patterns and similarities in the data and groups similar information together without being told how to group it.

Example:

A supermarket can use unsupervised learning to group customers based on their shopping habits. The model looks at how much customers spend and what products they buy, then groups customers with similar buying habits together.

This helps the supermarket understand its customers and create better marketing strategies.

B3. Differences Between Supervised and Unsupervised Learning

Comparison	Supervised Learning	Unsupervised Learning
Goal	To learn from known answers and make predictions.	To find hidden patterns and groups in data.
Type of Data	Uses labelled data with known answers.	Uses unlabelled data without known answers.
Typical Tasks	Classification and regression.	Clustering and dimensionality reduction.
Algorithms	Linear Regression, Logistic Regression and Decision Trees.	K-Means Clustering and PCA.
How Success Is Measured	By comparing predictions with actual answers using measures such as accuracy or mean absolute error.	By checking how well patterns or groups are formed, using measures such as silhouette score for clustering.
Real-World Example	A bank predicting whether a customer will repay a loan.	A supermarket grouping customers based on their shopping habits.

Summary:

The main difference is that supervised learning uses data with known answers to make predictions, while unsupervised learning uses data without known answers to discover patterns and similarities.

B4. Classification and Regression

1. Classification

Classification is a type of supervised learning that predicts a category or group. It uses labelled data to learn patterns and decide which category new information belongs to.

Example: A bank can use classification to predict whether a transaction is fraudulent or not fraudulent.

Type of Prediction: A category, such as Yes or No.

2. Regression

Regression is a type of supervised learning that predicts a numerical value. It learns from existing data to estimate a number.

Example: A property company can use regression to predict house prices based on the size, location and number of bedrooms.

Type of Prediction: A number, such as a house price of R850,000.

Difference:

Classification predicts categories, while regression predicts numerical values. Both use labelled data to learn and make predictions.

B5. Clustering and Dimensionality Reduction

1. Clustering

Clustering is a type of unsupervised learning that groups similar data together without using predefined labels.

Example: A supermarket can use clustering to group customers based on their spending habits. Customers who buy similar products or spend similar amounts can be placed in the same group.

Purpose: To find patterns and similarities in data.

2. Dimensionality Reduction

Dimensionality reduction is a technique used to reduce the number of features or columns in a dataset while keeping as much useful information as possible.

Example: A hospital may have patient information such as age, weight, blood pressure, blood sugar and other health measurements. Dimensionality reduction can combine related information into fewer features, making the data easier to analyze.

Purpose: To simplify large datasets, reduce unnecessary information and make data analysis easier.

Difference:

Clustering groups similar data together, while dimensionality reduction reduces the number of features in a dataset.

B6. Machine Learning Scenarios

a) Predicting a House's Selling Price

- Type: Supervised Learning
- Task: Regression

Reason: The model learns from previous house prices, sizes and suburbs to predict the selling price of a new house. It is regression because the predicted result is a number.

b) Grouping Customers by Spending Habits

- Type: Unsupervised Learning
- Task: Clustering

Reason: The model groups customers based on their spending habits without being given predefined groups. It is clustering because it finds customers with similar behaviour.

c) Deciding Whether an SMS Is Spam

- Type: Supervised Learning
- Task: Classification

Reason: The model learns from messages already labelled as spam or not spam. It is classification because it predicts which category a new message belongs to.

d) Predicting a Learner's Final Mark

- Type: Supervised Learning
- Task: Regression

Reason: The model uses previous learners' term marks and final marks to predict a learner's final result. It is regression because the predicted mark is a numerical value.

e) Finding Groups of Products With Similar Sales Patterns

- Type: Unsupervised Learning
- Task: Clustering

Reason: The model looks at product sales patterns and groups products with similar performance without predefined categories. It is clustering because it identifies similarities in the data.

B7. Scikit-learn Algorithms and Their Uses

1. Linear Regression

- Type: Supervised Learning
- Use: Predicts numerical values by finding a linear relationship between features and the target.
- Example: Predicting house prices based on house size.
- Module: `sklearn.linear_model.LinearRegression`

2. Logistic Regression

- Type: Supervised Learning
- Use: Predicts categories, such as Yes or No.
- Example: Predicting whether an email is spam or not spam.
- Module: `sklearn.linear_model.LogisticRegression`

3. Decision Tree Classifier

- Type: Supervised Learning
- Use: Makes classification decisions by splitting data into smaller groups based on different conditions.
- Example: Predicting whether a customer will repay a loan.
- Module: `sklearn.tree.DecisionTreeClassifier`

4. K-Neighbors Classifier

- Type: Supervised Learning
- Use: Classifies new data by looking at the categories of its nearest data points.
- Example: Predicting whether a patient has a disease based on similar patient records.
- Module: `sklearn.neighbors.KNeighborsClassifier`

5. K-Means

- Type: Unsupervised Learning
- Use: Groups similar data into clusters without using predefined labels.
- Example: Grouping customers based on their shopping habits.
- Module: `sklearn.cluster.KMeans`

6. PCA (Principal Component Analysis)

- Type: Unsupervised Learning
- Use: Reduces the number of features in a dataset while keeping as much important information as possible.
- Example: Reducing many patient health measurements into fewer features for easier analysis.
- Module: `sklearn.decomposition.PCA`

C1. Exploring and Cleaning a Dataset

Before using a dataset for machine learning, we need to understand its structure and check for problems such as missing values, duplicates and incorrect data types.

1. `df.info()`

Shows the column names, data types and number of non-missing values in a dataset.

2. `df.describe()`

Provides a statistical summary of numerical columns, including the mean, minimum, maximum and standard deviation.

3. `df.isnull().sum()`

Counts the missing values in each column so we can identify where data cleaning is needed.

4. `df.duplicated()`

Checks whether a dataset contains duplicate rows that may need to be removed.

5. `df.drop_duplicates()`

Removes duplicate rows from a dataset to avoid repeating the same information.

6. `df.astype()`

Changes a column from one data type to another, such as converting text into numbers.

Summary:

These functions help us understand the dataset, identify data quality problems and prepare cleaner data for machine learning.

C2. Handling Missing Values in a Dataset

Missing values are empty spaces in a dataset where information was not recorded. We need to handle them because they can affect the accuracy of a machine learning model.

1. `dropna()`

This function removes rows or columns that contain missing values. It is useful when only a small amount of data is missing and removing it will not affect the dataset too much.

2. `fillna()`

This function replaces missing values with another value, such as the mean, median, mode or a fixed value.

3. `Simple Imputer`

SimpleImputer is a scikit-learn tool that automatically fills missing values using a selected strategy. It is useful when preparing data for machine learning.

Ways to Replace Missing Values

- Mean: Uses the average value. It is suitable for numerical data without major outliers.
- Median: Uses the middle value. It is suitable for numerical data with outliers.
- Mode: Uses the most common value. It is useful for categorical data.

- Constant: Replaces missing values with a fixed value, such as "Unknown" for missing categories.

Summary:

The method used depends on the type of data and how much information is missing. Choosing the correct method helps improve data quality and makes the dataset ready for machine learning.

Example of Handling Missing Values

Suppose we have a dataset containing customers' ages:

Customer	Age
A	20
B	25
C	Missing
D	30

Solution:

We can replace the missing age with the mean (average) instead of deleting the customer's record.

Mean = (20 + 25 + 30) ÷ 3 = 25

Therefore, Customer C's missing age will be replaced with 25.

This helps us keep all customer records while handling the missing value.

C3. Encoding Categorical Data

Categorical encoding is the process of converting text or categories into numbers so that machine learning algorithms can understand and use the data.

1. Label Encoding

Label Encoding gives each category a different number.

Example:

- Red = 0

- Blue = 1
- Green = 2

It is commonly used to encode target labels in classification problems. For input features, the numbers may incorrectly suggest an order between categories.

2. Ordinal Encoding

Ordinal Encoding converts categories into numbers based on their natural order.

Example:

- Low = 0
- Medium = 1
- High = 2

It is suitable for ordinal data, where categories have a meaningful order.

3. One-Hot Encoding

One-Hot Encoding creates a separate column for each category and uses 0 or 1 to show whether the category is present.

Example:

Customer	Cash	Card	EFT
A (Cash)	1	0	0
B (Card)	0	1	0
C (EFT)	0	0	1

It is suitable for nominal data, where categories do not have a natural order.

4. Nominal vs Ordinal Data

- Nominal data: Categories without a natural order, such as gender, colour or payment method.
- Ordinal data: Categories with a meaningful order, such as low, medium and high.

5. Handling High Cardinality

High cardinality means a column contains many unique categories, such as hundreds of product names or customer cities.

Using One-Hot Encoding on such columns can create too many new features.

To handle this, we can group less common categories into an "Other" category or use techniques such as frequency encoding.

Summary:

The encoding method depends on the type of categorical data. One-Hot Encoding is useful for nominal data, Ordinal Encoding is suitable for ordered categories, and Label Encoding is commonly used for classification target labels.

C4. Feature Scaling

Feature scaling is the process of adjusting numerical values in a dataset so that features with larger numbers do not unfairly influence a machine learning model.

For example, a dataset may contain Age (18–65) and Salary (R5,000–R80,000). Scaling helps put these features on comparable scales.

1. Standardization (Z-score Scaling)

Standardization changes numerical values so that the data has a mean of 0 and a standard deviation of 1.

Formula:

Z = (X − Mean) ÷ Standard Deviation

It is useful for algorithms such as Logistic Regression, Support Vector Machines (SVM) and K-Nearest Neighbours (KNN).

2. Min-Max Scaling

Min-Max Scaling changes numerical values to a fixed range, usually between 0 and 1.

Formula:

X scaled = (X − Minimum) ÷ (Maximum − Minimum)

Example: If the minimum age is 20 and the maximum is 60, an age of 40 becomes:

(40 − 20) ÷ (60 − 20) = 0.5

This method is useful when we want numerical features to have the same range.

3. Robust Scaling

Robust Scaling uses the median and interquartile range (IQR) instead of the mean and standard deviation.

Formula:

X scaled = (X − Median) ÷ IQR

It is useful when a dataset contains outliers because extreme values have less influence on the scaling process.

4. Which Algorithms Need Scaling?

Algorithms that usually benefit from scaling include:

- K-Nearest Neighbours (KNN)
- Support Vector Machines (SVM)
- Logistic Regression
- Linear Regression, especially when regularization is used
- K-Means Clustering
- Principal Component Analysis (PCA)

Decision Trees and Random Forests generally do not need feature scaling because they split data based on feature values rather than distances.

Summary:

Feature scaling helps numerical features work together more fairly. Standardization uses the mean and standard deviation, Min-Max Scaling puts values into a fixed range, and Robust Scaling is useful when outliers are present.

C5. Train/Test Split and Data Leakage

1. What Is a Train/Test Split?

A train/test split is when we divide a dataset into two parts. The training data is used to teach the machine learning model, while the testing data is used to check how well the model performs on new information.

For example, we can use 80% of the data for training and 20% for testing.

2. What Is `train_test_split()`?

`train_test_split()` is a scikit-learn function that divides data into training and testing sets.

3. What Is `random_state`?

`random_state` controls the random splitting of data. Using the same number helps us get the same split when running the code again.

4. What Is `stratify`?

`stratify` helps keep similar proportions of each target category in both the training and testing datasets.

For example, if 70% of customers are labelled as non-defaulters and 30% as defaulters, stratification helps maintain these proportions in both sets.

5. What Is Data Leakage?

Data leakage happens when information that should not be available during training is used to train the model. This can make the model appear more accurate than it really is.

To prevent leakage, we should split the data before fitting preprocessing tools such as scalers or imputers. These tools should learn from the training data only.

6. Difference Between `fit()`, `transform()` and `fit_transform()`

- `fit()` learns the necessary information from the data, such as the mean and standard deviation.
- `transform()` applies the learned changes to the data.
- `fit_transform()` performs both steps together.

Summary:

Splitting data helps us evaluate how well a model performs on unseen information. We should fit preprocessing tools on the training data and then use them to transform the testing data to avoid data leakage.

Import the function used to split a dataset

from sklearn.model_selection import train_test_split

Split features (X) and target (y) into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(

X,

y,

test_size=0.2, # Use 20% for testing

random_state=42 # Make the split reproducible

)

C6. Detecting and Handling Outliers

Outliers are values that are very different from most other values in a dataset. They can affect data analysis and machine learning results.

1. Interquartile Range (IQR)

The IQR measures the difference between the first quartile (Q1) and third quartile (Q3).

IQR = Q3 − Q1

Values below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR are commonly identified as possible outliers.

2. Z-score

A Z-score measures how far a value is from the mean in terms of standard deviations.

Values with a Z-score greater than 3 or less than −3 are often considered possible outliers**.**

3. Boxplots

A boxplot is a chart that helps us see the spread of data and identify possible outliers. Outliers usually appear as individual points outside the whiskers.

Should Outliers Always Be Removed?

No, outliers should not always be removed. Some may be caused by data entry errors, while others may represent real and important information.

For example, a very high customer purchase amount could be a genuine transaction. We should investigate the value before deciding whether to remove it.

Summary:

IQR, Z-scores and boxplots help us identify unusual values. We should only remove or correct outliers when there is a valid reason.

C7. Feature Engineering

Feature engineering is the process of creating new columns or improving existing features in a dataset to help a machine learning model understand the data and make better predictions.

1. Creating a New Column

We can create a new column using information from existing columns.

Example: In a retail sales dataset, we can create a new column called Total Amount by multiplying Quantity by Price Per Unit.

Formula:

Total Amount = Quantity × Price Per Unit

For example, if a customer buys 3 products at R50 each:

Total Amount = 3 × R50 = R150

This new column helps us understand the total value of each transaction.

2. Using Correlation

Correlation helps us understand the relationship between numerical columns.

Example: We can check whether Quantity and Total Amount are related to understand how the number of products purchased is associated with transaction values.

Summary:

Feature engineering helps us create useful information from existing data. This can improve data analysis and help machine learning models identify patterns.

Part D.

D1. Overfitting and Underfitting in Machine Learning

Overfitting and underfitting are two problems that can affect how well a machine learning model makes predictions.

Overfitting happens when a model learns the training data too closely, including unnecessary details and noise. The model may perform very well on training data but struggle when given new data.

For example, a model predicting house prices might memorize the prices of houses in its training dataset instead of learning general patterns that apply to other houses.

Underfitting happens when a model is too simple to understand important patterns in the data. As a result, it performs poorly on both training and testing data.

For example, a house price model that only considers the number of bedrooms might miss important information such as location, size and property condition.

How to Prevent These Problems

Overfitting can be reduced by using simpler models, regularization, cross-validation or more representative training data.

Underfitting can be reduced by using a more suitable model, adding useful features or reducing excessive regularization.

Conclusion

The aim is to build a model that learns useful patterns without memorizing the training data. This helps the model make more reliable predictions on new information.

Reference:

Scikit-learn Developers. (n.d.). *Underfitting vs. overfitting*. https://scikit-learn.org/stable/auto_examples/model_selection/plot_underfitting_overfitting.html

D2. Model Evaluation Metrics in Machine Learning

Model evaluation metrics are used to measure how well a machine learning model performs. They help us understand whether a model is making accurate predictions and identify areas that need improvement.

For classification models, common evaluation metrics include accuracy, precision, recall and F1-score.

Accuracy measures the percentage of predictions that a model gets correct. However, accuracy can be misleading when one class appears much more frequently than another.

Precision measures how many of the predictions classified as positive are actually positive. It is useful when false positive predictions are costly.

Recall measures how many of the actual positive cases the model correctly identifies. It is important when missing a positive case could have serious consequences.

F1-score combines precision and recall into one measure. It is particularly useful when a dataset has imbalanced classes.

For regression models, metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) measure the difference between predicted and actual numerical values.

Conclusion

Choosing the correct evaluation metric depends on the type of machine learning problem and the consequences of incorrect predictions. Using appropriate metrics helps us understand model performance and make better decisions.

Reference:

Scikit-learn Developers. (n.d.). *Metrics and scoring: Quantifying the quality of predictions*. https://scikit-learn.org/stable/modules/model_evaluation.html

D3. Cross-Validation in Machine Learning

Cross-validation is a technique used in machine learning to evaluate how well a model is likely to perform on new, unseen data. It helps us check whether a model can make reliable predictions instead of performing well only on the data used during training.

One common method is K-Fold Cross-Validation. In this method, the dataset is divided into a specific number of groups called folds. For example, in 5-Fold Cross-Validation, the dataset is divided into five parts.

The model is trained five times. Each time, four folds are used for training, while the remaining fold is used for validation. This process continues until every fold has been used for validation once.

The evaluation results from all five rounds are then averaged to estimate the model's performance.

Benefits of Cross-Validation

Cross-validation helps identify overfitting, provides a more reliable performance estimate than a single training-validation split, and allows different machine learning models to be compared.

It is also useful when selecting model hyperparameters because it shows how different settings perform across multiple data splits.

Conclusion

Cross-validation is important because it helps us evaluate whether a machine learning model can generalize to unseen data. This supports better model selection and more reliable predictions.

Reference:

Scikit-learn Developers. (n.d.). *Cross-validation: Evaluating estimator performance*. https://scikit-learn.org/stable/modules/cross_validation.html

D4. Difference Between Parameters and Hyperparameters in Machine Learning

Parameters and hyperparameters are both important in machine learning because they influence how a model learns and makes predictions. However, they have different purposes.

Parameters are values that a machine learning model learns automatically from the training data. They help the model identify patterns and relationships between variables.

For example, in linear regression, the model learns coefficients and an intercept that determine how input features are used to predict an outcome. These values are adjusted during the training process.

Hyperparameters are settings chosen before training or selected through a tuning process. They control how the model is trained or how complex it can become.

For example, in a decision tree, the maximum tree depth is a hyperparameter. It controls how many levels the tree can have. In K-Nearest Neighbors, the number of neighbors is another example of a hyperparameter.

Why Are They Important?

Parameters allow the model to learn from data, while hyperparameters help control the learning process and model complexity.

Hyperparameter tuning involves testing different settings to find those that produce better results. Techniques such as Grid Search and Randomized Search can be used for this purpose.

Conclusion

Understanding the difference between parameters and hyperparameters is important because both influence model performance. Choosing suitable hyperparameters can help improve predictions and reduce overfitting or underfitting.

Reference:

Scikit-learn Developers. (n.d.). *Tuning the hyper-parameters of an estimator*. https://scikit-learn.org/stable/modules/grid_search.html

D9. Machine Learning Pipelines

A machine learning pipeline is a method of organizing different steps in a machine learning project into one structured process. These steps may include handling missing values, encoding categorical variables, scaling numerical features and training a model.

Pipelines are useful because they help make machine learning workflows more consistent, organized and easier to repeat.

For example, in my Shop Performance practical, I cleaned missing values, converted Payment Method into numerical features using One-Hot Encoding, and scaled Quantity and Discount using Standard Scaler. These steps could be combined into a machine learning pipeline instead of running them separately.

Benefits of Machine Learning Pipelines

One important benefit is that pipelines help prevent data leakage. When used correctly, preprocessing methods learn information from the training data without using information from the testing data.

Pipelines also reduce repeated code and make it easier to apply the same preprocessing steps to new data.

Another benefit is that pipelines can be used with cross-validation and hyperparameter tuning, making it easier to compare models and improve their performance.

Conclusion

Machine learning pipelines are important because they simplify data preprocessing and model training. They help create more reliable, consistent and reusable machine learning workflows.

Reference:

Scikit-learn Developers. (n.d.). *Pipeline: Chaining estimators*. https://scikit-learn.org/stable/modules/compose.html#pipeline-chaining-estimators

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Scikit-learn Developers. (n.d.). *Tuning the hyper-parameters of an estimator*. https://scikit-learn.org/stable/modules/grid_search.html

Scikit-learn Developers. (n.d.). *Pipeline: Chaining estimators*. https://scikit-learn.org/stable/modules/compose.html#pipeline-chaining-estimators

AI-Use Declaration

I used ChatGPT as a support tool while completing my Machine Learning Research Assignment.

ChatGPT helped me understand machine learning concepts, clarify explanations and organize my research. It also provided guidance on Python coding and data preprocessing.

I used Databricks to run my Python code, examine the results and complete the practical section. I also used Canva to prepare and present my research report.

AI was used as a learning and guidance tool throughout the assignment.

AI Tool Used: ChatGPT (OpenAI)

Purpose: Learning support, research assistance and coding guidance.